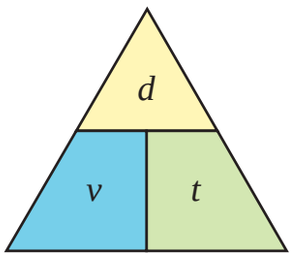
Average velocity = Total displacement/Time taken.
(Taking the direction due east of point O as positive)

Average velocity = (5 – 7) / 0.2
= –2 / 0.2
= –10 km/h or – 10 × 5/18
= – 25/9 = –0.28 m/s

The triangle method can help you to recall the relationship between velocity (v), displacement (d), and time(t).



v = d / t , t = d / v, d = v × t

Answer the following questions.

- Calculate the velocity of a car travelling with a uniform velocity covering 100 m in 4 seconds.
- Usain Bolt covers 100 m in 9.58 seconds. Calculate his speed. If Usain Bolt competes with a Cheetah which is running at a speed of 30 m/s, who will be the winner?
- You are walking along east direction covering a distance of 4 m, then 2 m towards south, then 4 m towards west and at last 2 m towards north. You cover the total distance in 21 seconds. What is your average speed and average velocity?

2.3 Acceleration

Acceleration is the rate of change of velocity. In other words, if a body changes its speed or direction then it is said to be accelerated.

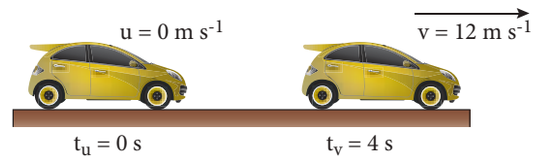
$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time}}$$

$$= \frac{\text{Final velocity (v)} - \text{Initial velocity (u)}}{\text{Time}}$$

$$a = \frac{(v - u)}{t}$$

SI unit of acceleration is m/s^2

A car at rest starts to travel in a straight line path. It reaches a velocity of 12 m/s in 4 s . What is its acceleration, assuming that it accelerates uniformly?



Initial velocity, u = 0 m/s (Since the car starts from rest)
Final velocity (v) = 12 m /s
Time taken (t) = 4 s

$$\text{Acceleration (a)} = \frac{(v - u)}{t} = \frac{(12 - 0)}{4} = 3\,m/s^2$$



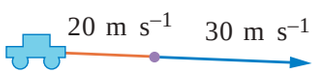
See how brisk I am !



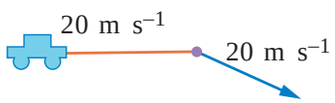
My name is cheetah. I can run at great speed. Do you know what my speed is? It is 25 m/s to 30 m/s. My speed changes from 0 to 20 m/s in 2 second. See how good my acceleration is ! Can you calculate it?

2.3.1 Positive acceleration

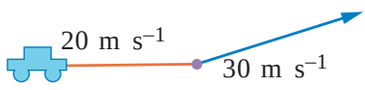
If the velocity of an object increases with respect to time, then the object is said to be in positive acceleration.



(a) Change in speed



(b) Change in direction



(c) Change in both speed and direction

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